Whenever you are ready let’s go: we will reuse an algorithm that we have seen in the previous chapters, specifically a helper method we used in clustering.

Remember k-means[6]? We had this problem of initializing the centroids: come up with an initial choice for k points in a n-dimensional space. It looked easier than it actually was, didn’t it? And yet we saw how important a good choice for the initial points was, to get a fast convergence and a good result.

Take a look at section 12.2, and in particular at listings 12.1 and 12.2, to see how we implemented the random initialization of points for the centroids.

Here, with the graph embedding, we have a similar problem: we need to choose a certain number of 2D points, and the way we chose them will directly determine the result (there isn’t, this time, an optimization algorithm running after the choice).

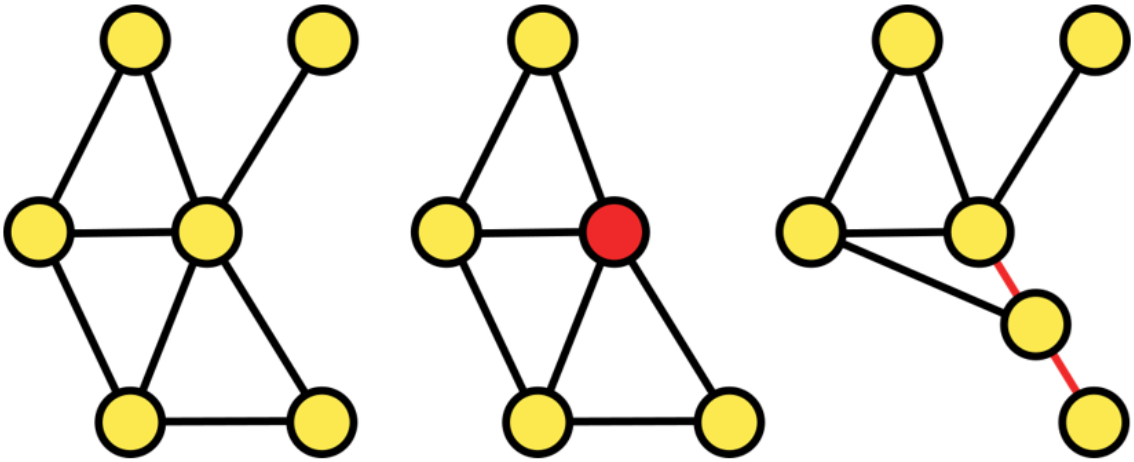


Figure 16.2 Three embeddings for the same graph. On the left, a valid embedding. The embedding in the middle is not valid because two vertices are assigned the same position, while the one on the right has a vertex intersecting an edge (one for which it’s not an endpoint).